Menakan Thamilchelvan

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Education

McMaster University — Hamilton, Ontario
Bachelor of Engineering in Mechatronics Engineering (CO-OP), GPA: 3.7 — *Sep. 2023 – Apr. 2028*
Coursework: Data Structures & Algorithms, Embedded Systems, Dynamic Systems & Control, Signals & Systems

Experience

Linamar Corporation (Robotics & Automation) — Sep. 2026 – Present
AI / Software Engineer Co-op — *Guelph, Ontario*

- Developing and deploying in-house **robotic bin-picking cells**, integrating industrial 2D/3D depth cameras and fine-tuned **YOLO object detection** models to execute real-time part localization and 6-DoF pose estimation.
- Constructed high-fidelity digital twin environments in **NVIDIA Isaac Sim** to simulate bin-picking cells, photorealistic sensor noise, and physics-based contact dynamics for **reinforcement learning** policy training.
- Architected custom **Copilot AI agents** to automate controls engineering workflows, reducing routine PLC/robot telemetry logging and diagnostic scripting time by 35%.
- Prototyped **Physical AI and imitation learning pipelines** (Diffusion Policy, LeRobot) to collect multi-modal teleoperation datasets and benchmark adaptive robotic manipulation in high-mix environments.

Skyjack Inc. — Jan. 2026 – Aug. 2026
Design and Validation Engineer Co-Op — *Guelph, Ontario*

- Conducted evaluation and duty-cycle testing across **4+ mobile aerial platforms** using the **eDAQ-lite DAQ**, analyzing over **50+ GB** of multi-channel strain, pressure, and cycle test data.
- **Automated test data processing** and database entry by developing custom **Excel VBA macros**, reducing recurring engineering workload by **1 hour/day (∼5 hours/week)** and eliminating manual transcription errors.
- Managed the Testing and Validation **Master Schedule**, tracking **15+ concurrent machine test phases** and coordinating with cross-functional teams to achieve on-time launch milestones.
- Validated high-precision **GNSS RTK sensors** achieving **<2 cm positioning accuracy** across **30+ field trials**; integrated feeds with eDAQ-lite systems to preserve telemetry in RF-obstructed industrial environments.

McMaster Formula Electric — Feb. 2025 – Present
Low Voltage and Wire Harness Sub-team Member — *Hamilton, Ontario*

- Designed and routed a **2-layer PCB for an LED driver** using **Altium Designer**; minimized high-current switching loops and return paths to suppress EMI under a 1.5A peak load.
- Designed and crimped **low-voltage harness assemblies** and verified continuity; maintained wiring diagrams and BOM.

Projects

Custom Smartphone Gimbal for Stabilization | *Inventor, ESP32, 3D Printing, C++*

- Designed and implemented a custom two-axis smartphone gimbal to stabilize video recording; reduced visible shake by 60% on walking tests.
- Developed firmware in C++ integrating **MPU6050 sensors with ESP32**; PID control loop at 200 Hz with end-to-end latency under 10 ms.
- Utilized Solidworks to create lightweight CAD models for a printable frame; total assembly mass under 250 grams and material cost under $5.
- Powered the system with a custom battery pack and buck-boost converter; achieved more than 3 hours of continuous operation in field use.

Technical Skills

Languages & Frameworks: C/C++, Python, ROS 2 (Humble), PyTorch, OpenCV, Visual Basic (VBA)
Robotics & Physical AI: NVIDIA Isaac Sim, Digital Twins, Reinforcement Learning (RL), Diffusion Policy, LeRobot, YOLO11
Hardware & Embedded: Industrial 2D/3D Cameras, ESP32, eDAQ-lite DAQ, CAN Bus, GNSS RTK, MPU6050 IMU, Altium Designer
Software & Lab Tools: Linux (Ubuntu), Git, SolidWorks, Autodesk Inventor, MATLAB, LaTeX, 3D Printing, Harness Crimping, Soldering